

AGCL: Aspect Graph Construction and Learning for Aspect-level Sentiment Classification

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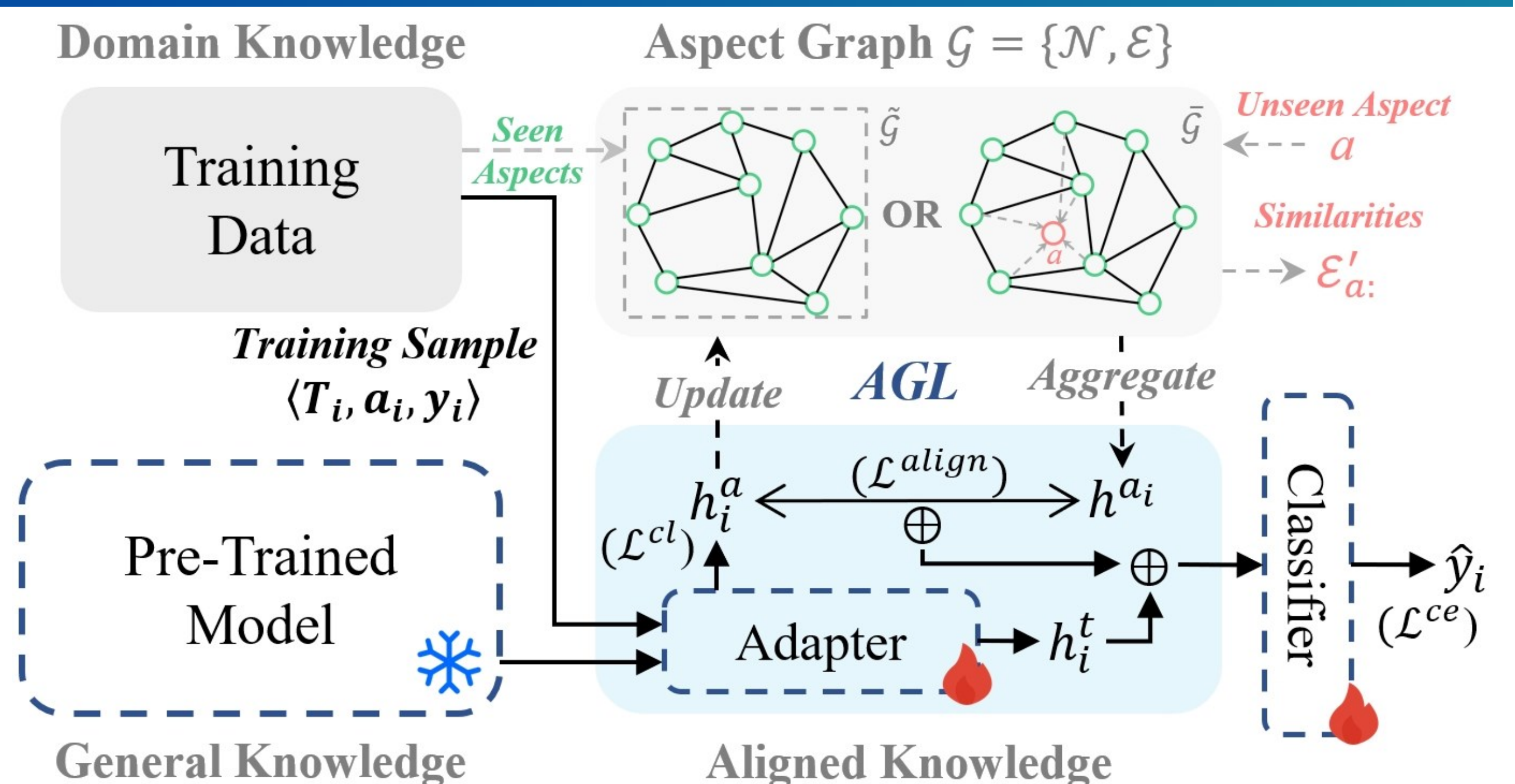


<https://github.com/jian-projects/agcl>

Abstract

Prior studies on Aspect-level Sentiment Classification (ALSC) emphasize modeling interrelationships among aspects and contexts but overlook the crucial role of aspects themselves as essential domain knowledge. To this end, we propose **AGCL**, a novel Aspect Graph Construction and Learning method, aimed at furnishing the model with finely tuned aspect information to bolster its task-understanding ability. AGCL's pivotal innovations reside in Aspect Graph Construction (**AGC**) and Aspect Graph Learning (**AGL**), where AGC harnesses intrinsic aspect connections to construct the domain aspect graph, and then AGL iteratively updates the introduced aspect graph to enhance its domain expertise, making it more suitable for the ALSC task. Hence, this domain aspect graph can serve as a bridge connecting unseen aspects with seen aspects, thereby enhancing the model's generalization capability. Experiment results on three widely used datasets demonstrate the significance of aspect information for ALSC and highlight AGL's superiority in aspect learning, surpassing state-of-the-art baselines greatly.

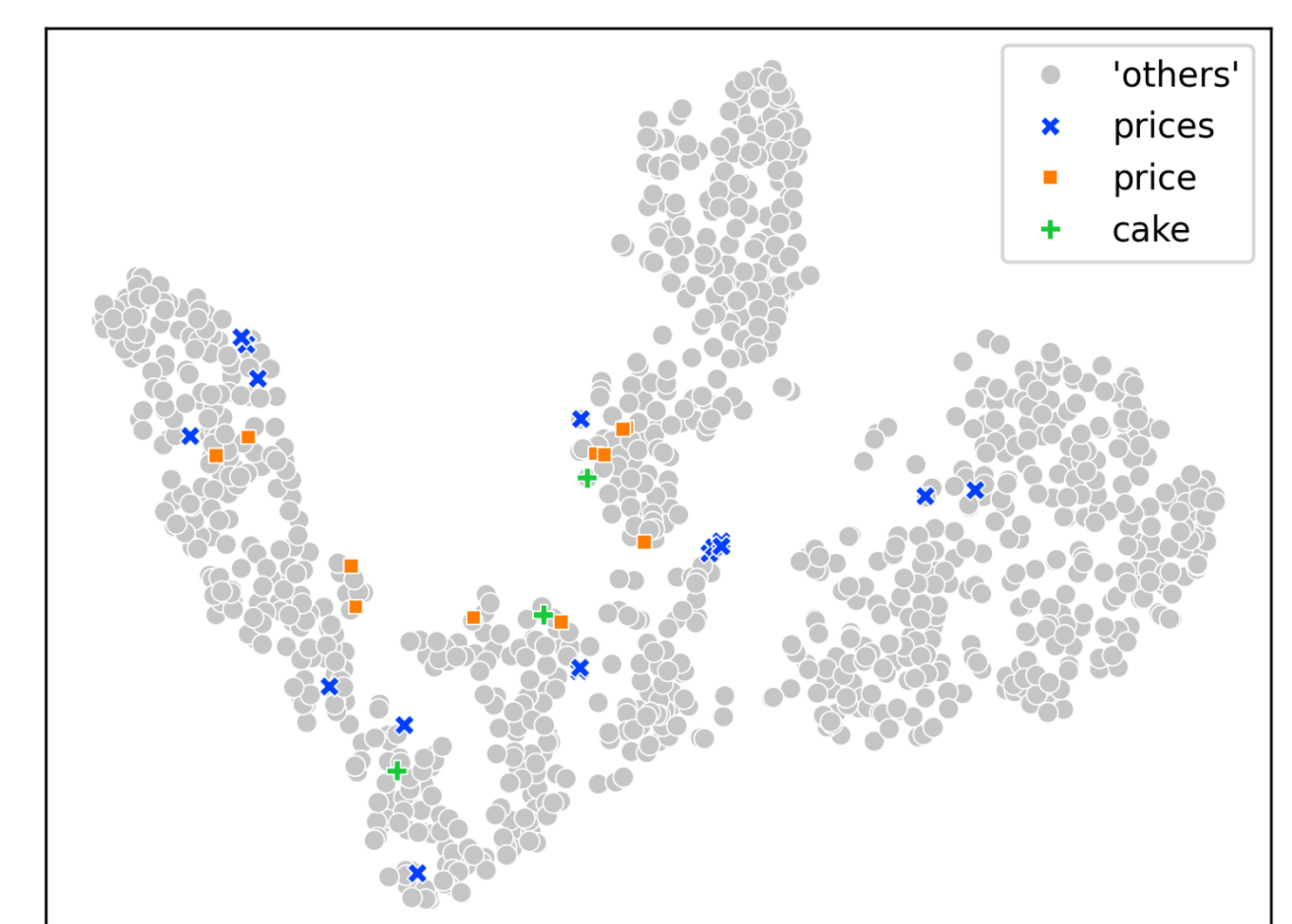
Method



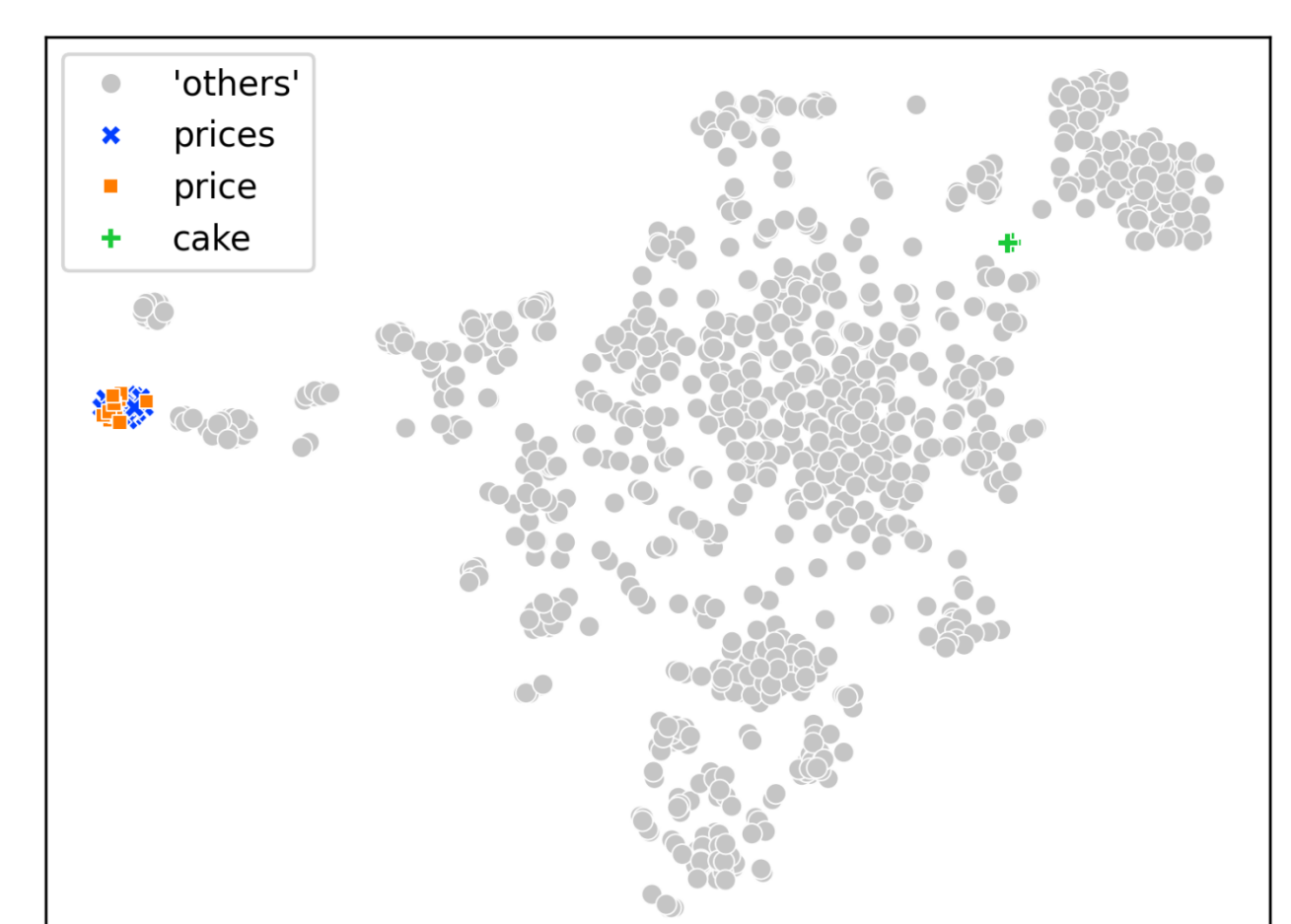
- ◆ **Aspect Graph Construction:** Given the expenses of building the aspect graph by domain experts, we suggest treating language models as experts to automate the Aspect Graph Construction.
- ◆ **Aspect Graph Learning:** To bolster the efficacy of the aspect graph, we develop the Aspect Graph Learning, including Enhance Aspect Representation, Update Aspect Graph, and Aspect Representation Rectification.

Experimental Results

Categories	Models	Laptops		Restaurants		Twitter	
		Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy
Prompt	GPT-3.5-turbo	73.16	80.00	75.17	87.15	61.67	60.96
Structure	DualGCN	78.10	81.80	81.16	87.13	76.02	77.40
	dotGCN	78.10	81.03	80.49	86.16	77.00	78.11
	BiSyn-GAT	79.15	82.44	81.63	87.49	76.80	77.99
	RoBERTa4GCN	78.16	81.80	78.61	86.23	74.00	74.75
	TextGT+BERT	78.71	81.33	82.27	87.31	76.45	77.70
Knowledge	♠BERTABSA	77.88	81.38	80.49	86.61	75.67	76.59
	BERTABSA-ATT	79.30	82.64	82.34	87.86	76.45	77.60
	ABSA-DeBERTa	79.36	82.76	83.42	<u>89.46</u>	-	-
	ABSA-ESA	79.34	82.44	81.74	88.29	-	-
	DR-BERT	78.16	81.45	82.31	87.72	76.10	77.24
	DeBERTa+RCL	80.28	82.76	<u>84.68</u>	89.38	77.47	78.32
	PConvRoBERTa	<u>80.89</u>	<u>83.54</u>	84.27	89.29	<u>77.53</u>	<u>78.47</u>
AGL (Ours)		82.15 _{↑1.26}	84.54 _{↑1.00}	85.63 _{↑0.95}	90.30 _{↑0.84}	78.15 _{↑0.62}	78.85 _{↑0.38}

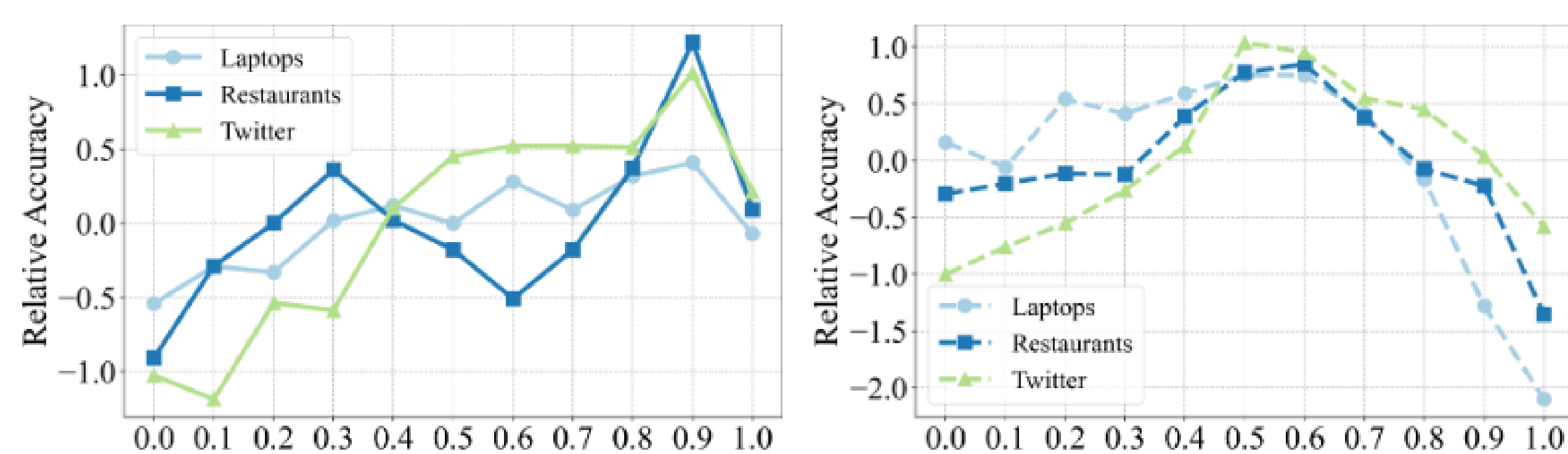


Model w/o AGL



Model w/ AGL

➤ Parameter Analysis of AGC and AGL



(a) ϵ (in Equation (8))

(b) λ (in Equation (6))

➤ Ablation Results on Restaurants

Methods	Macro-F1	Accuracy
AGL	85.63	90.30
w/o $[\mathcal{L}^{cl}]$	84.50 _{↓1.13}	89.76 _{↓0.54}
w/o $[\mathcal{L}^{align}]$	85.33 _{↓0.30}	90.06 _{↓0.24}
w/o $[\mathcal{L}^{cl}, \mathcal{L}^{align}]$	84.17 _{↓1.46}	89.34 _{↓0.96}

Conclusion

In this paper, we explore the role of aspect for ALSC instead of meticulous model component design. To this end, we propose AGCL, a novel method that constructs the domain aspect graph and employs it as expert knowledge to enhance the aspect representations. The key points of AGCL reside in two perspectives: 1) employing language models as experts to automatically build the domain aspect graph, and 2) developing effective mechanisms to update the aspect graph and facilitate the knowledge alignment process. Experimental results and ablation studies demonstrate the effectiveness of AGL in enhancing the aspect representations and improving the model's generalization ability. In the future, we plan to explore better ways to use the well-refined aspect information in the domain aspect graph.